

Munaza Ashraf

PhD Applicant in World Models, Embodied AI and Efficient Learning

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Education

Sep 2020 **B.Sc. (Hons.) Software Engineering, University of Engineering and Technology (UET)**
Jun 2024 *Taxila, Pakistan*

- CGPA **3.94/4.0**, **Silver Medalist** (second highest aggregate across all four years).
- **Final Year Project: PsycheSync**, an AI mental health platform with mood tracking, journaling, assessments, and a moderated peer community (Flutter).
- Fine tuned **LLaMA 2** on medical guidance for precautions, remedies, medicines, prescriptions, and meditation, and fine tuned **OpenAI** model weights to role play as a psychiatrist.

Research Interests

Areas Machine Learning, Multimodal and Generative AI, Foundation and World Models, Computer Vision and 3D Vision, Embodied AI and Robot Learning, Efficient Model Training, Distillation and Inference

Direction Developing efficient **world and multimodal models for embodied and physical AI**, with a focus on transferring spatial, temporal, and physical reasoning into compact models through **knowledge distillation** and efficient training and inference under strict latency, memory, and compute constraints.

Publications & Preprints

2026 **Recovering a Withheld Audio Tokenizer from Its Generator: Reverse Distillation of the MiniMax Music 3 Encoder, In and Out of Distribution**
Under Review

Munaza Ashraf, bghira. *Under review*, 2026. [\[Code\]](#) [\[Models\]](#) [\[Dataset\]](#)

- Recovered the unreleased audio to code tokenizer of MiniMax Music 3 (8 residual quantised code streams) by **reverse distillation** from the generator's own codes and code distributions.
- Showed **out of distribution transfer**: recorded music recovers as large a share of reachable conditioning as generated music (**0.92 vs. 0.91**), and unseen genres are encoded as well as seen ones.
- Identified code collapse under band limiting and lossy coding; degraded view training repairs it (4 kHz low pass: **0.82 vs. 0.50**) at a cost of only **0.01** on clean audio.
- A decoder causal across codebook depth adds **+0.10** condition replay cosine, while parameter matched controls gain nothing.
- Released encoders, code, and **MM3 OOD**, a new **240** track test set over **12** held out genres; all models retrained under **3** seeds.

Independent Research & Reproductions

Aug 2026 **World to Policy Transfer via Online World Model Distillation, WPT, [Code]**

Sep 2026 **Tools:** OccWorld, nuScenes, PyTorch

- Independently implemented and open sourced WPT for autonomous driving on a frozen **OccWorld** world model.
- Built the full loop: world model rollouts, teacher trajectory proposal, reward scoring, and **policy distillation** into a compact student.

May 2026 **Head Wise Adaptive Sparse Attention for Video Diffusion, HASTE**

Tools: Wan 2.2, LTX 2.3, Sparse Attention

- Implemented training free video diffusion acceleration with temporal mask reuse and error guided budgeted calibration using per head top p thresholds.

- 2025 **Diffusion Inference Caching Benchmark**
- 2026 **Tools:** FBCache, TeaCache, MagCache, Cache DiT, Fast3D Cache, Nsight Systems
- Benchmarked caching methods across image, video, and 3D generative models, measuring kernel time and peak VRAM against FID, CLIP score, SSIM, and LPIPS.
- May 2026 **Efficient Fine Tuning and Optimizer Studies**
- Jul 2026 **Tools:** TREAD, DreamBooth, LoRA, ConsisLoRA, SOAP, Muon, AdamW, Lion
- Studied **TREAD token routing** with LoRA and context parallelism, measuring convergence, memory, throughput, and quality.
 - Ran controlled **optimizer comparisons** (SOAP, Muon, AdamW, Lion) for diffusion, LLM, and VLM adapters.
 - Designed **segmented gradient checkpointing** for Flux and MageFlow for a finer memory and recompute trade off.

Research and Work Experience

- Nov 2024 **Applied ML Engineer, Runware, London, UK (Remote)**
- Present **Tools:** PyTorch, Diffusers, Transformers, vLLM, Go, CUDA, ROCm, NCCL, FlashAttention, SageAttention, FlashInfer, Docker, Prometheus, Vercel
- Integrated, optimized, and served **1,000+** image, video, audio, 3D, LLM/VLM, and multimodal models in production, with up to **2×** faster inference on selected workloads.
 - Built model specific schedulers and execution recipes for **FLUX, ACE Step, TRELIS 3D, Ideogram, Anima, MiniMax, and LTX**, covering batching, VRAM aware placement, and hot swapping.
 - Designed **distributed and serverless inference** across NVIDIA RTX, Hopper, and Blackwell GPUs (RTX PRO 6000, H100, H200, B300) and AMD ROCm, using context parallelism with NCCL communication overlap.
 - Benchmarked FlashAttention, SageAttention, FlashInfer, and cuDNN SDPA per GPU architecture to route each model to its **fastest attention backend**, combined with diffusion and KV caching and **FP8, FP4, INT8, INT4** quantization.
 - Worked on **teacher student knowledge distillation**, and deployed single LoRAs across base, distilled, and turbo variants via per layer representation variance analysis.
 - Served high volume enterprise workloads for **Higgsfield, Freepik, Wix, PixVerse, Civitai, and the LTX team**, with API and frontend integrations deployed on Vercel.
- Sep 2024 **Founding Engineer, MedTalk, Faz Australia, Australia (Remote)**
- Jul 2025 **Tools:** Whisper, AssemblyAI, MedPaLM, LLaMA, RAG, Pinecone, Vector Search, WebSockets, AWS, RunPod
- Founding engineer for **MedTalk**, a real time clinical AI platform used by **50,000+** clinicians; owned ML, backend, and production ML infrastructure.
 - Curated and prepared clinical training datasets, and fine tuned speech, medical NLP, and language models including **MedPaLM** for structured clinical note generation.
 - Cut post VAD streaming speech latency from **~150ms to ~50ms P50 (3×** faster) on Whisper and then AssemblyAI, with WebSocket streaming and optimized model serving.
 - Built a **RAG based clinical assistant** on Pinecone vector search, grounding LLM answers in retrieved medical context.
 - Built a **retrieval augmented billing code LLM** that maps consultations to medical billing codes, deployed on AWS.
 - Implemented privacy and consent controls for HIPAA, Australian Privacy Principles, and PIPEDA.
- Jun 2024 **Computer Vision Engineer, VisionRD, Islamabad, Pakistan**
- Nov 2024 **Tools:** YOLO, OpenCV, PyTorch, Real Time Video Pipelines
- Built a **wearable AI assembly copilot** for Toyota (head mounted 4K camera, real time error detection, cycle analysis), improving line efficiency by **~35%**.
 - Built automated worker activity and assembly validation systems for Thal Engineering.
 - Developed low latency visual defect detection for KIA Motors to automate quality inspection.

- 2022 **STEM Trainer, Junior Coderz, Islamabad, Pakistan (Remote)**
- 2024 **Tools:** Scratch, Python, App Development, Robotics
- Taught live classes in Scratch, Python, app development, and robotics to students across **Canada, the USA, and the UAE**.
 - Coached students preparing for **USACO** and **Bebras** computing competitions in the USA, UK, and Europe.
 - Mentored AI projects and designed project based lessons in computational thinking.
- Mar 2023 **Machine Learning Intern, NESCOM, Islamabad, Pakistan**
- Oct 2023
- Developed a deep learning ad detection and classification system, covering data collection, preprocessing, feature engineering, and evaluation.
 - Worked on signal processing and pattern recognition with a focus on accuracy.
- May 2022 **Deep Learning Intern, Bytewise Limited, Islamabad, Pakistan**
- Sep 2022
- Designed and trained computer vision and NLP models in TensorFlow and PyTorch.
 - Built custom datasets, augmentation pipelines, and classification and detection workflows.

Selected Projects

- 2026 **Juricratic: AI Litigation Intelligence, Personal Venture, with Connor, juricratic.com**
- Present
- A litigation intelligence layer that turns fragmented case data into strategic foresight.
- Models legal matters as **imperfect information games**, running thousands of reproducible simulations to map the space of case outcomes before motions are filed.
 - Synthetic Bench of four role bound **LLM agents** (judge, jury, opposing counsel, client) that stress test arguments, attack surfaces, and settlement positions.
 - Matter state inference ingests new facts in plain English and propagates their impact across claims, with source linked assertions, confidence scores, and attorney review gates.
- Nov 2024 **Multimodal Model Inference & Optimization Platform, Runware, runware.ai**
- Jul 2026
- Built a unified execution layer for image, video, audio, 3D, and LLM/VLM models with routing, batching, streaming, hot swapping, and VRAM aware scheduling.
 - Integrated caching, sparse and parallel attention, compilation, and automated profiling to compare latency, throughput, VRAM, and output quality across models and hardware.
- May 2026 **Multi Architecture LoRA Training Platform, Runware, runware.ai/lora-training**
- Jul 2026
- Engineered modular LoRA and QLoRA training for diffusion, LLM, and VLM architectures with dataset preprocessing, automatic captioning, checkpointing, and validation.
 - Implemented adapter evaluation, dynamic LoRA loading, merging, weighted composition, and production hot swapping.
 - Used **4 bit and 8 bit** quantization, mixed precision, and gradient checkpointing; swept LoRA rank, alpha, target modules, learning rate, and optimizers.
- Apr 2026 **vLLM Microservice Platform in Go, Runware**
- May 2026
- Built a model agnostic Go microservice layer with JSON APIs for prompt enhancement, captioning, age detection, and content safety scoring.
 - Wrote inference recipes for FLUX, WAN, LTX, MiniMax, Klein, and Anima with KV caching, compilation, and GPU and host memory management.
 - Containerized with Docker, published authenticated ReDoc docs via Cloudflare, and ran **LLMOps** with Prometheus monitoring.
- Mar 2026 **Padelytix: Real Time Multi Camera Sports Perception**
- Aug 2026
- Video analytics pipeline for player and ball tracking, clothing based reidentification, court homography, and highlight extraction.
 - Focused on robust perception under occlusion, identity persistence, camera timing mismatch, and fast post match inference.
- Jul 2025 **Automated Lumber Takeoff and Cost Estimation from Construction Drawings**
- Feb 2026
- Built a multi stage pipeline with YOLO detection, SAM segmentation, and Detectron2 region analysis to extract lumber types, dimensions, quantities, and structural context.
 - Integrated Gemma and Qwen vision language models to interpret drawing annotations, deployed as a FastAPI cloud service for automated bills of materials and cost estimates.
- Apr 2025 **Unilever × PSL 2025: Broadcast Logo Detection and Sponsorship Analytics**
- Jun 2025
- Built a real time YOLOv8 and OpenCV pipeline to detect, track, and grade sponsor logo visibility while measuring screen time across live broadcast footage.
 - Aggregated detections with match context (overs, camera angles, events) into automated exposure reports and ROI based sponsorship analytics.

Honors and Awards

- 2024 **Silver Medal**, *UET Taxila*, second highest aggregate in the session across all four years
- 2022 to 2023 **Laptop Award**, *Prime Minister's Youth Laptop Scheme*, Pakistan
- 2019 **1st Position in Pre Engineering**, *Year 12 (HSSC)*, college wide
- 2018 to 2020 **Full Tuition Fee Waiver and Stipend**, *Years 11 and 12 (HSSC)*, awarded on Year 10 (Matriculation) results

Technical Community & Leadership

- 2020 **Active Member**, *Google Developer Student Club (GDSC)*, UET Taxila
- 2024
 - Organised workshops and led hands on sessions on TensorFlow, Firebase, and Flutter.
- 2020 **Active Member**, *Microsoft Learn Student Ambassadors (MLSA)*
- 2024
 - Co organised talks and study jams on Azure and ML, and mentored peers on deploying ML projects.
- 2020 **Event Host**, *IEEE, UET Taxila Chapter*
- 2024
 - Hosted seminars and panels connecting students with industry professionals in AI and software.
- 2020 **General Secretary**, *Umeed-e-Subh*, UET Taxila
- 2024
 - Student run welfare community that established a school for underprivileged children across from UET Taxila.
 - Helped run daily classes and an evening IT drive teaching graphic design, basic coding, and video editing so children could earn while still in school.

Relevant Coursework

- Mathematics** Calculus, Linear Algebra and Differential Equations, Probability and Statistics
- ML & Vision** Machine Learning, Neural Networks, Computer Vision, Digital Image Processing
- Systems** Data Structures and Algorithms, Operating Systems, Databases, Cloud Computing

Technical Skills

- Programming** Python, C++, CUDA, Go, SQL, Unix Shell
- ML** PyTorch, TensorFlow, Diffusers, Transformers, PEFT, TorchAO, LoRA and QLoRA, Diffusion and Flow Models, LLMs, VLMs, Knowledge Distillation
- Train & Infer** FSDP, DTensor, Tensor and Context Parallelism, Quantization (FP8, FP4, INT8, INT4), Mixed Precision, FlashAttention, SageAttention, Ring and Sparse Attention, KV Caching, Diffusion Caching, PyTorch Compilation
- GPU Systems** NVIDIA RTX, Hopper, Blackwell; AMD ROCm; Triton, CUTLASS, cuDNN, NCCL
- Infrastructure** Distributed and Serverless Inference, Go Microservices, vLLM, LLMOps, FastAPI, Docker, Kubernetes, Prometheus, AWS, RunPod, Vercel
- Vision** YOLO, SAM, Detectron2, Tracking, Reidentification, Pose Estimation
- Evaluation** Nsight Systems and Compute, Latency and Throughput Benchmarking, FID, CLIP Score, SSIM, PSNR, LPIPS

References

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